TOWARDS TRANSPARENT EMBEDDINGS FOR BIOMARKER DISCOVERY

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ABSTRACT

Foundation models often yield highly predictive embeddings; however, their semantic content can be opaque, which limits real-world use. We introduce BioLENS, a framework for ordered-outcome settings that models latent structure as a smooth outcome–embedding trajectory g, then extracts function-level features (e.g., trend, curvature, extrema) to summarize how representations vary across outcome levels. In a biomarker-based minimum viable prototype (MVP), we instantiate g using embeddings of neural time series with a smoothing spline, and translate clinically motivated structure into explicit constraints to evaluate whether latent geometry aligns with meaningful outcome progression. More generally, the MVP illustrates how domain-informed desiderata can be translated into quantitative, testable constraints for comparing encoders, and how implicit representation learning objectives can incorporate sparse clinical labels alongside large unlabeled datasets to encourage embeddings that are not only predictive but also outcome-relevant and interpretable.

1 INTRODUCTION

The interpretability challenge for latent spaces. In high-stakes domains such as finance, medicine, and law, it is often not enough for a model to perform well. Stakeholders also need principled ways to understand what drives its outputs and when those outputs should be trusted. Traditional interpretability tools such as feature importance, permutation tests, and Shapley-style attributions (Breiman, 2001; Fisher et al., 2019; Lundberg & Lee, 2017) are effective when model inputs correspond to meaningful, human-defined variables; however, when features are opaque, entangled, or lack clear semantics, attribution becomes difficult to translate into actionable explanations.

Such limitations have become increasingly salient amid the rapid proliferation of foundation models in recent years (Devlin et al., 2019; Brown et al., 2020). Foundation models are typically trained at scale to produce general-purpose latent representations (embeddings) that compactly encode increasingly rich and complex input information in a fixed-dimensional latent space (Bengio et al., 2012). In practice, these embeddings are often consumed primarily as a black-box basis for downstream fine-tuning rather than as objects of scientific interpretation (Bommasani et al., 2021).

Motivation: neurobiomarker discovery. The limited semantics of embeddings is especially consequential in neuroscience and neurotechnology, where raw measurements are high-dimensional, noisy, and mechanistically complex. Electroencephalography (EEG) measures electrical activity recorded non-invasively from scalp electrodes, often at high sampling rates and across multiple channels. This makes it a natural target for representation learning, yet a difficult substrate for interpretation (Biasiucci et al., 2019).

Recent work suggests that large-scale self-supervised pretraining can learn representations from EEG that transfer across tasks and cohorts, improving performance in clinically relevant settings (Chau et al., 2025). Yet, the limited interpretability of these representations constrains their adoption for applications that demand transparent clinical reasoning, including diagnosis, stratification, and treatment planning, even when the representations appear to capture highly relevant information. Existing attempts to understand foundation model embeddings, such as correlating embedding coordinates with handcrafted signal features or metadata, linear probing models, and

clustering-based analyses provide partial insight (Kim et al., 2018; Alain & Bengio, 2016; van der Maaten & Hinton, 2008). However, they rarely yield explanations that connect quantitative embedding structure to human-interpretable concepts.

A promising framework: TIMEVIEW. To move beyond correlational and probing-style analyses of embeddings, we draw inspiration from recent interpretability advances in time series forecasting. Specifically, Kacprzyk et al. (2024) present TIMEVIEW: a "bi-level transparency" framework that formalizes how humans naturally describe trajectories in terms of their trends (global shape) and the properties of those trends (e.g., minima, maxima), rather than as collections of pointwise values. Motivated by this top-down view of trajectory comprehension, interpretability is framed as understanding how perturbations of the input change the overall trend of the predicted trajectory (level 1) and the key properties of that trend (level 2). This approach contrasts bottom-up feature attribution approaches that operate over individual prediction horizons or pre-specified time points.

Contributions. Our contributions are as follows:

1. We introduce BioLENS, a framework for interpreting embeddings in ordered-outcome settings motivated by biomarker discovery.

2. BioLENS yields human-interpretable summaries of how representations change across outcome levels and uses them to derive testable, domain-informed criteria for candidate embeddings.

3. We instantiate BioLENS in an EEG-based minimum viable prototype (MVP), illustrating how domain expectations can guide both embedding evaluation and representation learning objectives beyond downstream performance.

2 BioLENS

BioLENS (Biomarker-discovery-driven Latent Embedding iNterpretation via Splines) is a TIMEVIEW-inspired framework for interpreting embeddings by encoding structured relationships between a latent space and an ordered outcome. Given outcome–embedding pairs $\{(y_i, z_i)\}$, where $z_i \in \mathbb{R}^k$ is a k-dimensional embedding vector and y_i is ordered (e.g., an ordinal label, risk score, or outcome quantile), we model the embedding as a continuous trajectory g indexed by y. We estimate g with a smooth fit, which makes it straightforward to extract derivative-based, human-interpretable structure.

Analogous to TIMEVIEW's decomposition of trajectories into locally consistent motifs separated by transition points, BioLENS uses the geometry of g to produce interpretable statements about trend direction, curvature, and qualitatively meaningful changes as the outcome varies. This yields a representation-level analogue of TIMEVIEW's composition map: a mapping from outcome intervals and corresponding embeddings to descriptions of how the representation evolves.

Crucially, BioLENS is not only post hoc: it provides a way to translate domain expectations (e.g., "risk should rise sharply over a clinically relevant interval, then saturate") into explicit constraints on g (e.g., positive slope over an interval, concave-down curvature, transition points near clinically meaningful bounds). These constraints can be used to evaluate whether an embedding supports a target use case and to define objectives/regularizers for learning embeddings that exhibit desired structure beyond downstream task performance.

Figure 1 summarizes BioLENS end-to-end. The top row depicts the generic pipeline (Section 3), while the shaded bottom row shows the MVP instantiation (Section 4).

3 Formulation

We first formulate the problem generically. Section 4 then outlines an EEG-based MVP.

3.1 Setup and Assumptions

Let $X = \{x_1, x_2, \ldots, x_n\}$ denote inputs and let $E : X \to \mathbb{R}^k$ be a fixed encoder that maps each x_i to an embedding $z_i = E(x_i) \in \mathbb{R}^k$. Let $Y = \{y_1, y_2, \ldots, y_n\}$ be the corresponding outcomes. We assume outcomes lie on a one-dimensional ordered domain $I \subset R$, with $y_i \in I$.

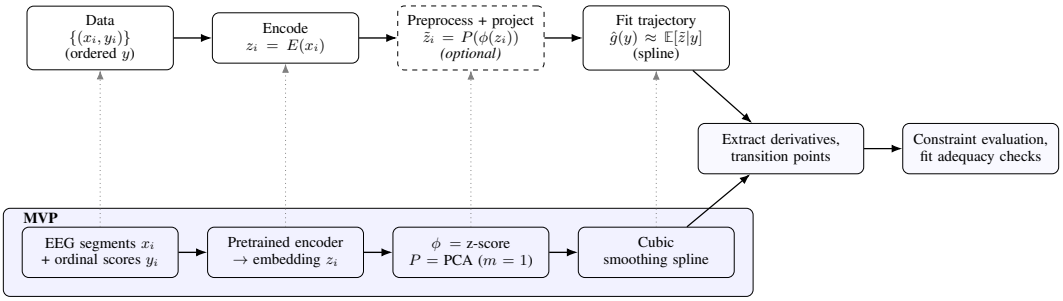

Figure 1: BioLENS algorithm flow: generic pipeline (top) with an MVP instantiation (bottom).

3.2 Optional Preprocessing and Projection

Optionally, we apply a preprocessing map $\phi : \mathbb{R}^k \to \mathbb{R}^k$ (e.g., normalization) followed by a projection $P : \mathbb{R}^k \to \mathbb{R}^m$, yielding projected embeddings

$$\tilde{z}_i = P(\phi(z_i)) \in \mathbb{R}^m.$$

When omitted, $\tilde{z}_i = z_i$ and $m = k$.

3.3 Trajectory Estimation

We seek a smooth outcome–embedding trajectory $g : I \to \mathbb{R}^m$ with $g \in C^2(I)$ (i.e., g is at least twice differentiable) that summarizes how embeddings vary with the ordered outcome. We parameterize g as a degree-d vector-valued B-spline:

$$g(y) = \sum_{j=1}^{p} c_j \, B_{j,d}(y)$$

where $B_{j,d}$ are B-spline basis functions of degree d and $c_j \in \mathbb{R}^m$ are coefficient vectors. For an open/unclamped B-spline construction, the total number of basis functions is $p = K + d + 1$ where K is the number of interior knots.

Estimation procedure. We form the dataset $D = \{(y_i, \tilde{z}_i)\}_{i=1}^n$ and fit the spline parameters $\{c_j\}$ to approximate $\tilde{z}_i \approx g(y_i)$. We then compute characteristics of g along I (e.g., transition points) to obtain human-interpretable structure relating embeddings to outcomes.

4 Minimum Viable Prototype: EEG Biomarker Discovery

We now instantiate the generic formulation from Section 3 in the context of EEG-based biomarker discovery. We extend the framework by formalizing domain-informed desiderata as testable constraints on the trajectory g, yielding quantitative criteria for evaluating candidate embeddings.

4.1 Formulation

Data. Let $D = \{(x_i, y_i)\}_{i=1}^n$ where $x_i \in X \colon \mathbb{R}^{C \times T}$ is an EEG recording with C channels and T time steps, and $y_i \in I \subset \mathbb{R}$ is an ordered outcome (e.g., cognitive score or symptom severity). Each pair (x_i, y_i) comes from the same recording/assessment instance (e.g., a task session), with y_i derived from the associated assessment.

4.2 INSTANTIATING BIOLENS

Encoder and embeddings. We use a pretrained encoder E that maps each recording to an embedding $z_i = E(x_i) \in \mathbb{R}^k$. We consider single-channel recordings sampled at 128 Hz over 10 minutes ($C = 1, T = 128 \times 10 \times 60 = 76800$) and an embedding dimension $k = 10$.

Preprocessing and projection. We apply z-scoring followed by a 1D PCA projection $P : \mathbb{R}^{10} \to \mathbb{R}$, yielding scalar embeddings $\tilde{z}_i \in \mathbb{R}$. We use PCA1 only as a convenient 1D readout for the MVP; our framework applies to any chosen (or learned) scalarization for post hoc trajectory fitting (see Section 4.4)

Trajectory estimation. We fit a cubic smoothing spline with $K = 5$ interior knots to obtain a 1D trajectory $g : I \to \mathbb{R}$ relating the projected embedding coordinate $\tilde{z}$ to the ordered outcome y.

Generic and MVP-specific parameter values are summarized in Table 1.

Table 1: BioLENS parameters and MVP instantiation.

	Parameter	Description	MVP settings
Formulation	k	embedding dimension of encoder E	10
	ϕ	preprocessing operator	z-score
	P	projection operator	PCA1
	m	projection output dimension; spline vector dimension	1 (scalar)
	d	degree of spline g	3
	K	number of interior knots of spline g	5
	p	number of basis functions in spline g	9
MVP Dataset	C	number of channels in EEG signal $x_i \in X : \mathbb{R}^{C \times T}$	1
	T	number of time steps in EEG signal $x_i \in X : \mathbb{R}^{C \times T}$	76800

4.3 BIOMARKER DISCOVERY CONSTRAINTS

Clinical relevance of trajectory shape. Modeling the candidate embedding as a smooth function of outcome y makes it possible to define and evaluate clinically relevant shape properties of the outcome–embedding relationship. This mirrors how biomarker evidence is often grounded in qualitative curve structure: for example, Colagiuri et al. (2011) model retinopathy prevalence—a diabetes-defining microvascular endpoint—as a function of several glycemic measures (FPG, 2-h PG, and HbA1c). By examining the onset/acceleration region of these curves (Figure 2), they identify glycemic ranges where risk meaningfully increases, providing empirical support for diagnostic threshold ranges and candidate cutpoints.

Desiderata as constraints. Building on this perspective, we formalize three clinically motivated desiderata as constraints on the fitted trajectory $g(y)$:

1. **Proximity.** Transition points occur near clinically significant boundaries.
2. **Monotonicity.** Over clinically significant ranges, the trajectory has a consistent slope direction.
3. **Reproducibility.** These properties are stable across cohorts and under resampling.

Below, we formalize each constraint as a quantitative, testable condition on $g(y)$.

Constraint 1: Proximity. Define the point-to-set distance $d(y, T) := \min_{t \in T} |y - t|$. We say g satisfies *proximity* to $[y_1, y_2]$ if

$$d(y_1, T(g)) \leq \tau_1 \quad \text{and} \quad d(y_2, T(g)) \leq \tau_2$$

for predefined τ_1 and τ_2.

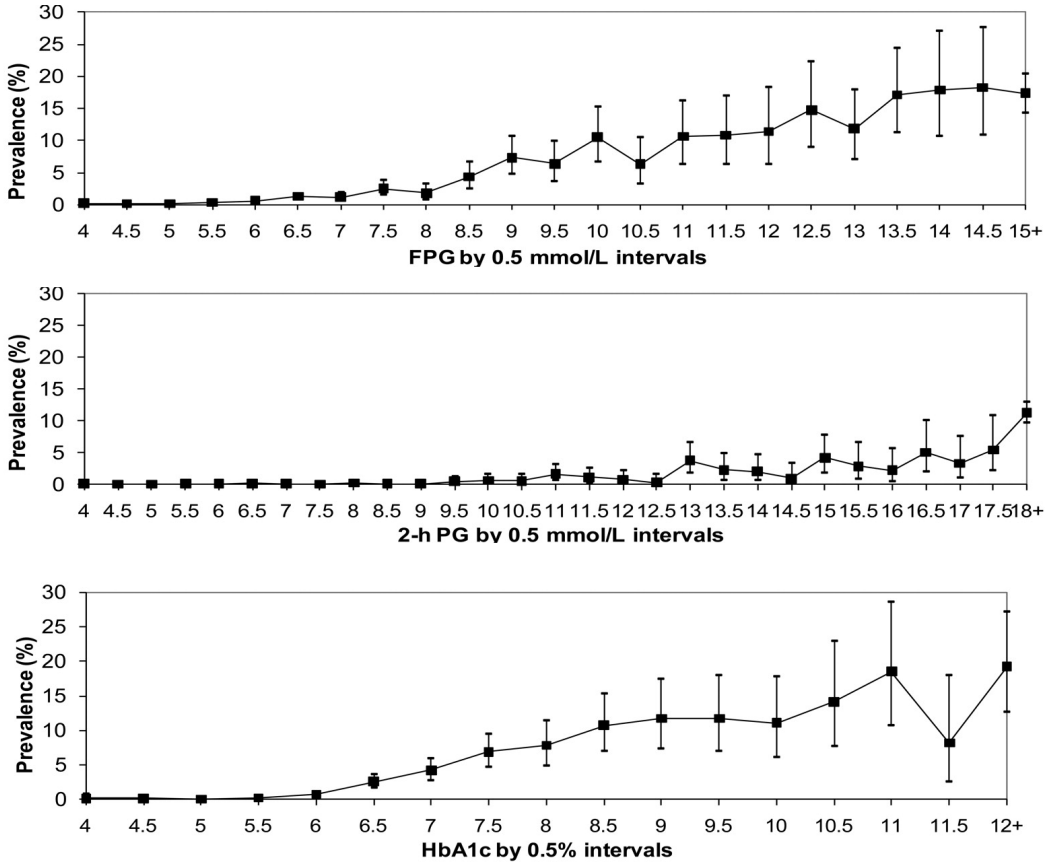

Figure 2: Prevalence of diabetes-specific retinopathy versus fasting plasma glucose (FPG; top), 2-h post oral glucose load plasma glucose (2-h PG; middle), and hemoglobin A1c (HbA1c; bottom), shown in fixed-width bins with 95% confidence intervals. The curves exhibit a localized onset/acceleration region, largely sustained slope direction, and stability of these qualitative features under sampling variability (reflected by the confidence intervals), motivating the constraints outlined in Section 4.3. Adapted from Colagiuri et al. (2011).

Constraint 2: Monotonicity. Let $I := [y_1, y_2]$. Define the total variation of g over I as $\mathrm{TV}(g; I) := \int_{y_1}^{y_2} |g'(y)|\, dy$, and the net change $\Delta g := g(y_2) - g(y_1)$. We define

$$M_{\mathrm{tv}}(g; I) := \frac{|\Delta g|}{\mathrm{TV}(g; I)},$$

and say g satisfies *monotonicity* on I if $M_{\mathrm{tv}}(g; I) \geq \rho$ for predefined $\rho \in [0, 1]$. While the Spearman rank correlation between y and the scalar embedding coordinate is a reasonable global measure of monotonic association, it neither supports restriction to a clinically motivated sub-interval nor leverages the smooth derivative structure of g—both of which are central to our trajectory-level interpretability criteria.

Constraint 3: Reproducibility. Let $C(D)$ be any boolean-valued condition evaluated on dataset D. For bootstrap resamples $D^{(1)}, \ldots, D^{(B)}$, define

$$\widehat{R} := \frac{1}{B} \sum_{b=1}^{B} \mathbb{1}\{C(D^{(b)})\}.$$

We say C satisfies *reproducibility* if $\widehat{R} \geq \alpha$, i.e., the condition holds in at least an α fraction of bootstrap resamples. In practice, we apply reproducibility to the proximity and monotonicity conditions.

Fit adequacy (precondition). In addition to satisfying constraints 1-3, the fitted $g(y)$ should be sufficiently accurate such that derivative-based characteristics are meaningful. We note two potential checks:

- **Comparison to null distribution.** Compute mean squared error (MSE) on a holdout set. Generate a null distribution by repeatedly shuffling y, refitting $g(y)$, and computing holdout MSE. Compare the true holdout MSE to this null distribution (e.g., via a one-sided permutation p-value) (Ojala & Garriga, 2010). A statistically significant improvement over the null indicates that the observed fit is unlikely to arise from chance alignment between outcome labels and embeddings, and therefore supports the claim that the embedding contains information about y.
- **Improvement relative to baseline.** Compute holdout MSE of $g(y)$ and compare to a baseline predictor (e.g., training set mean prediction, linear regression). An improvement in MSE indicates that the outcome–embedding relationship is not only non-random, but also that the spline captures structure beyond what is explained by a simpler relationship; in particular, improvement over a linear baseline suggests the presence of meaningful nonlinear (yet smooth) dependence of the embedding on y.

An example of a viable and non-viable biomarker assessed per these constraints can be found in Figure 3.

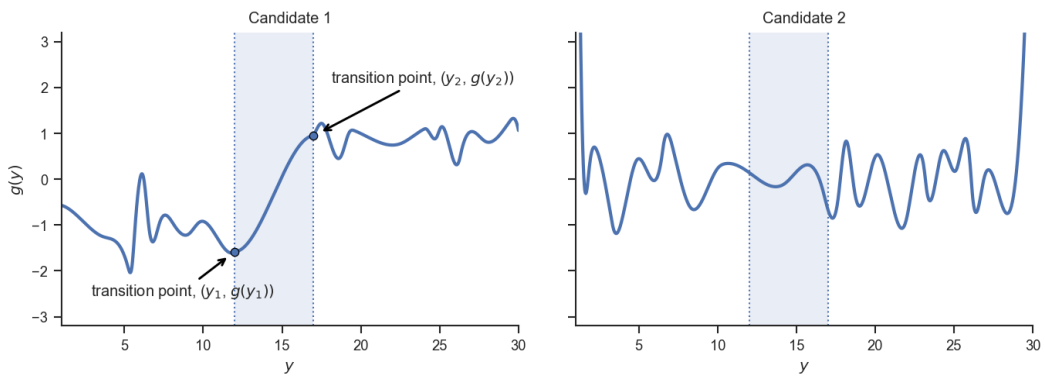

Figure 3: Viable vs. non-viable biomarker examples. Candidate 1 (left) satisfies proximity and monotonicity in $[y_1, y_2]$; Candidate 2 (right) does not.

4.4 Constraint-aware learning

In settings where large unlabeled datasets are available but clinical outcome labels are sparse—a common scenario in neuroscience and medicine—the same constraints in Section 4.3 can be used not only for post hoc analysis, but also as implicit objectives that encourage the learning of outcome-relevant representations during training. We illustrate this idea by deriving a differentiable loss that encourages the monotonicity constraint (Constraint 2) and integrating it with a self-supervised learning (SSL) framework.

4.4.1 Monotonicity regularization

Let E_θ be a trainable encoder with embeddings $z_i = E_\theta(x_i) \in \mathbb{R}^k$. Since monotonicity in Section 4.3 is defined over a scalar trajectory $g : I \to \mathbb{R}$, we introduce a scalar readout

$$s_i = h_{\text{mono}}(\phi(z_i)) \in \mathbb{R},$$

where ϕ denotes any preprocessing used in the pipeline (e.g., z-scoring). We treat s_i as the one-dimensional embedding coordinate used to fit $g(y)$.

For any labeled pair (i, j), a monotone increase with respect to y implies

$$y_i > y_j \Rightarrow s_i \geq s_j, \qquad \text{equivalently} \qquad (y_i - y_j)(s_i - s_j) \geq 0.$$

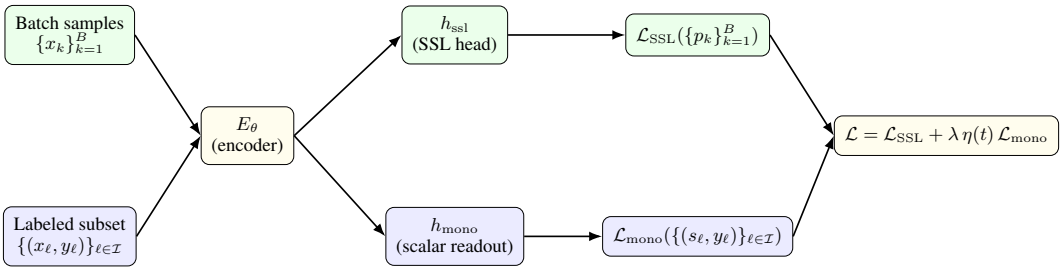

Figure 4: Constraint-aware training with a shared encoder and two heads. The SSL objective $\mathcal{L}_{\text{SSL}}$ is computed over all B samples in the batch, while the monotonicity term $\mathcal{L}_{\text{mono}}$ is computed only over the labeled subset $\mathcal{I}$ within the batch and weighted by $\lambda\,\eta(t)$. Both terms update E_θ.

We penalize violations with the smooth surrogate

$$\mathcal{L}_{\text{mono}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \text{softplus}(-\beta(y_i - y_j)(s_i - s_j))\,,$$

where $\mathcal{P}$ denotes a set of labeled pairs (e.g., pairs within a batch, optionally restricted or weighted to a clinically relevant interval I) and $\beta > 0$ the sharpness of the penalty. This formulation promotes monotone increases; to target monotone decreases, one can flip the sign of the pairwise term.

4.4.2 Joint training with self-supervision.

To preserve the scalability and transfer benefits of self-supervised learning while injecting lightweight clinical structure from labeled examples, we combine $\mathcal{L}_{\text{mono}}$ with a standard self-supervised objective $\mathcal{L}_{\text{SSL}}$ (e.g., contrastive or masked reconstruction) computed over both labeled and unlabeled samples:

$$\mathcal{L} = \mathcal{L}_{\text{SSL}} + \lambda\,\eta(t)\,\mathcal{L}_{\text{mono}}, \qquad \lambda \geq 0,\ \eta(t) \in [0, 1].$$

Here, λ trades off interpretability-constrained structure against generic representation learning, and $\eta(t)$ is a warm-start schedule that increases over training. Warm-starting allows the model to first establish a stable latent space under $\mathcal{L}_{\text{SSL}}$ (when the outcome–embedding relationship may be noisy or poorly shaped early in training), then gradually enforce monotonic structure as representations stabilize. As illustrated in Figure 4, $\mathcal{L}_{\text{SSL}}$ operates over the full batch to leverage unlabeled data for general representation learning, while $\mathcal{L}_{\text{mono}}$ uses only the labeled subset to inject sparse clinical structure.

4.4.3 Practical considerations

Gradient stability and relative magnitude. In realistic settings, only a small fraction of a batch may be labeled with y, which can make the monotonicity term $\mathcal{L}_{\text{mono}}$ sparse and noisy (or even undefined when too few labeled examples are present). This is exacerbated by the fact that $\mathcal{L}_{\text{SSL}}$ is computed over all samples and can dominate the optimization unless the weighting schedule $\eta(t)$ is tuned carefully. In practice, we recommend (i) enforcing a minimum number of labeled samples per batch and (ii) warming up $\lambda(t)$ from near-zero to a target value.

Which coordinate to interpret. Monotonicity is explicitly imposed on the learned scalar readout $h_{\text{mono}}(\phi(z)))$, not on an unsupervised axis such as PCA1. Consequently, post hoc interpretability analyses should fit the spline trajectory $g(y)$ directly to s rather than PCA1; otherwise, the monotone structure may not be visible due to arbitrary rotations in the embedding space.

Pair selection and label noise. The monotonicity loss depends on labeled pair construction. Using all labeled pairs can overweight densely sampled regions of y and amplify noise for near-tied outcomes. We therefore suggest (i) restricting to pairs with $|y_i - y_j| \geq \epsilon$ (or weighting pairs by $|y_i - y_j|$), and (ii) optionally subsampling pairs (e.g., selecting pairs that are adjacent in y) to control variance and computation.

5 Related Works

Here, we explain how our work intersects with related areas of ML and EEG-based biomarker discovery.

Foundation model embedding interpretability. Most embedding interpretability work for foundation models is post hoc. For example, linear probes and diagnostic classifiers test what is decodable from frozen representations (Kim et al., 2018), while concept-based methods quantify the influence of human-defined directions on predictions (Koh et al., 2020). A complementary line of work notes that interpretability can be a coordinate artifact; for example, axis-aligned semantics may disappear under rotations even though the model's behavior is unchanged (Bau et al., 2017). In contrast, BioLENS is not primarily a probe: it treats embeddings as substrates for constraint-driven, trajectory-level explanations by explicitly modeling a smooth outcome–embedding curve and evaluating it for domain-informed properties, making interpretability a target rather than a retrospective diagnostic.

EEG-based biomarker discovery. EEG-based biomarker discovery has traditionally focused on interpretable feature families (bandpower, event-related potentials, connectivity) (Biasiucci et al., 2019), but these are often fragile to preprocessing, cohort heterogeneity, and other confounding factors (Soares de Aguiar Neto & Garcia Rosa, 2019). Recent work explores latent spectral descriptors that track symptom dimensions (Hacker et al., 2025), while self-supervised EEG representations often remain weakly linked to clinically actionable biomarkers despite effective transferability (Kostas et al., 2021; Chau et al., 2025). BioLENS aims to close this gap by converting pretrained embeddings into explicit, testable biomarker hypotheses informed by domain expectations rather than relying only on downstream task performance.

Implicit representation learning objectives. Self-supervised learning makes clear that objectives define latent geometry: contrastive instance discrimination (Chen et al., 2020; He et al., 2020) and masked reconstruction (He et al., 2022) yield different invariances, while time series SSL further injects temporal assumptions via neighborhood or multi-scale objectives (Tonekaboni et al., 2021; Yue et al., 2022). Separately, shape-constrained modeling shows how domain priors can be enforced directly through differentiable architectures/regularizers (You et al., 2017). BioLENS is novel in combining these ideas: it operationalizes interpretability as an implicit learning objective on the representation itself, creating a pathway from generic representation learning to embeddings whose geometry is deliberately aligned with plausible outcome progression.

6 Limitations and Future Work

BioLENS makes interpretability concrete by fitting a smooth outcome–embedding trajectory g, aligning with TIMEVIEW's premise that humans reason about trajectory structure through trends and their properties. While this yields a clean interface for derivative-based statements, several limitations remain and motivate future work.

Projection and scalarization. The MVP projects embeddings to a single scalar coordinate prior to spline fitting. While convenient, collapsing a k-dimensional embedding to a scalar can discard clinically relevant structure and interactions among latent factors, and may obscure outcome-dependent variation that is not well-aligned with the chosen or learned axis (Shlens, 2014). As a result, trajectory fits and constraint scores can understate (or miss) structure that is present in the full embedding space.

Dimensionality and tractable interpolation. Although the generic formulation supports vector-valued trajectories, scaling interpolation and interpretation to higher-dimensional latent spaces quickly becomes challenging, requiring increasing amounts of data to resolve local behavior and decreasing the extent of interpretability. Low-rank spline coefficients and related approaches could preserve more information while keeping the trend abstraction usable.

Thresholding and domain knowledge. BioLENS relies on thresholds τ_1, τ_2, ρ, and α for proximity, monotonicity, and reproducibility constraints. These may be brittle across cohorts and outcomes. By incorporating domain knowledge as priors or explicit utility objectives, the selection of these thresholds could be made more principled and portable.

7 Conclusion

We introduced BioLENS, a TIMEVIEW-inspired framework for interpreting embeddings in ordered-outcome settings by modeling representations as a smooth outcome–embedding trajectory g and extracting derivative-based structure to produce human-interpretable summaries. In our EEG-based MVP, we instantiated g using standardized embeddings, a low-dimensional projection, and a smoothing spline. We then explored how the resulting trajectory behavior could be used to formulate mathematical characteristics of viable biomarkers: proximity (whether key transitions occur near clinically meaningful thresholds), monotonicity (whether behavior is consistent over outcome ranges of interest), reproducibility (whether these properties persist under bootstrap resampling), and fit adequacy checks to guard against over-interpreting noise or poorly fit trajectories. Finally, we extended our approach from post hoc analysis to constraint-aware learning by deriving a monotonicity regularizer and demonstrating how it can be combined with standard self-supervised objectives—operating over large unlabeled datasets while using sparse clinical labels for structural guidance—to encourage outcome-aligned latent structure during training.

Our MVP is best understood as one instance of the broader principle that, by leveraging a BioLENS-like approach, domain-informed desiderata can be formalized as explicit, testable constraints on the relationship between an ordered outcome and candidate embedding. Beyond explanation, BioLENS offers a practical route to selecting, and ultimately learning, representations whose latent geometry is not only predictive but also aligned with plausible outcome structure, enabling a clearer bridge from foundation model embeddings to testable hypotheses.

Code availability statement. A reference implementation of the MVP, demonstrated on synthetic data, is available at `https://github.com/jacksampiere/biolens`.

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